

Brandon Y. Perez Martinez

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PROFESSIONAL SUMMARY

Self taught AI and software engineer with 6 years of U.S. Air Force cyber operations experience and 4+ years building production agentic AI systems and quantitative engineering platforms. Builds multi agent orchestration systems, custom MCP tooling, and fine tuned LLM pipelines, alongside full stack and cloud native applications shipped to real users and paying clients. DoD network operations background with 2 overseas deployments and an active CompTIA Security+.

TECHNICAL SKILLS

AI / Agentic Engineering: Multi agent orchestration across mixed model fleets (Claude, Codex, ChatGPT, Grok, Antigravity), Model Context Protocol (MCP) server development, LLM fine tuning (DPO/SFT) on open models (Qwen3-14B), local inference (Intel Arc), secure execution sandboxing (Kata Containers / microVMs for untrusted agent workloads), eval and trajectory analysis, retrieval augmented generation (RAG, QMD), prompt & tool calling design

Languages: Python (advanced), TypeScript, Bash, SQL, PowerShell

Data & ML: PostgreSQL, TimescaleDB, LightGBM, scikit-learn, walk forward validation, feature engineering, time series pipelines, ETL, REST API integration

Full Stack & Cloud: Next.js, React, Express, Drizzle ORM, Stripe, Cloudflare (Pages/Workers), Google Cloud Run, GCS, Docker, CI/CD (GitHub Actions, CodeQL, OpenSSF Scorecard)

Infra & DevOps: Linux (RHEL/Ubuntu/Debian), VPS fleet operations, systemd, cron automation, monitoring/failover, Git

Cybersecurity: Network security hardening, ACAS/Nessus, Active Directory, VMware/vSphere/ESXi, DoD network operations

EXPERIENCE

AI & Software Engineer | Independent / Self Employed | Remote | Aug 2022 - Present

- Built a multi agent AI orchestration system coordinating mixed fleets of Claude, Codex, ChatGPT, Grok, and Antigravity agents (planner + specialized sub agents) for autonomous software engineering, research, and operations, using custom MCP servers exposing tools to the models, with fail closed permission gating.
- Implemented and maintained retrieval augmented generation (RAG) pipelines with QMD over a personal knowledge base, giving agents grounded document retrieval and search inside day to day workflows.
- Designed a secure execution layer running untrusted, agent generated code no egress inside Kata Containers microVMs, with a fail closed admission gate: a production sandboxing pattern used by AI infra companies.
- Designed and deployed a production algorithmic trading platform (Python, PostgreSQL/TimescaleDB): 80+ live data streams, 26+ concurrent strategies, regime adaptive and Bayesian Kelly position sizing, a dedicated risk engine (circuit breakers, VaR/CVaR, correlation sizing), and genetic algorithm strategy weight evolution; execution across Hyperliquid, Kalshi, Polymarket, Coinbase, and Alpaca (live on Kalshi prediction markets, gated paper execution elsewhere). 1,100+ Python modules, 2,690+ commits, CI gated.
- Built an ML signal pipeline (LightGBM ensemble, 742K+ candles, 45 features, walk forward validation) with automated model lifecycle management.
- Operate full Linux VPS infrastructure: systemd services, 150+ cron jobs, real time monitoring, automated failover and alerting.
- Built large scale web data extraction and enrichment pipelines (headless browser automation, reverse engineered private/hidden APIs, Chrome DevTools Protocol) feeding a 100K+ record B2B database across 9 verticals with multi signal scoring; delivered as qualified B2B lead lists for client Diaz Solutions.

Full Stack Engineer | Greenlightdate (greenlightdate.com) | Client Engagement, then Co-owner (15%) | 2026

- Engaged as a client engineer to overhaul the existing codebase: refactored the application into a pnpm monorepo (Vite/React + Express + Drizzle/Postgres) with Stripe billing and OpenAPI → typed client codegen.
- Executed a live cloud migration (Replit → Vercel + Cloud Run → Cloudflare Pages + GCS) with zero downtime cutover.
- Joined the team after delivery as co-owner (15%) and sole backend engineer; hardened CI with CodeQL, dependency review, and OpenSSF Scorecard, and delivered the application through a security audit and compliance review.

Cyber Systems Operations Technician | U.S. Air Force, 52nd Combat Communications Squadron | Aug 2016 - Jul 2022

- Owned the server stack for deployable Command Post communications: built, fielded, and administered \$7.2M in Theater Deployable Communications flyaway kits, centered on the Flexible Communications Package (FCP), a self contained deployable data center (4 ESXi hosts, 2 switches, 2TB of RAM, 40TB of storage, and onboard UPS power) that stands up a complete network anywhere in the world on rapid deployment timelines.
- Managed 13 virtualized servers (VMware/vSphere/ESXi) end to end on the FCP, hosting the entire deployed enterprise for its users: Active Directory, Exchange email, file share, firewall and proxy, ACAS/Nessus vulnerability scanning, HBSS endpoint security, WSUS patching, alerting, backup, ticketing, and network monitoring; handled provisioning, patching, backup, and restoration for the whole stack.
- Led ACAS/Nessus vulnerability scanning and remediation programs; hardened Active Directory and DoD network infrastructure across classified and unclassified networks.
- **Syria (Operation Inherent Resolve, Oct 2018 - Feb 2019):** Restored 3,700+ mission critical files after a system failure under deployed combat conditions, preserving continuity of operations.
- **Niger (Operation Juniper Shield, Aug 2020 - Mar 2021):** Delivered a \$9.6M network upgrade and eliminated 3,000+ vulnerabilities across 16 domains, hardening 104 virtualized systems.
- Senior Airman (SrA, E-4) at separation after 6 years of service.

SELECTED PROJECTS

Multi Market Algorithmic Trading Platform: Production Python system trading crypto perps, prediction markets (Kalshi/Polymarket), and equities on one unified PostgreSQL/TimescaleDB pipeline. Risk engine (circuit breakers, VaR/CVaR, correlation sizing), genetic algorithm strategy evolution, LightGBM signal ensemble; 1,100+ Python modules, 2,690+ commits, CI gated, VPS deployed with live execution on Kalshi and gated paper execution elsewhere.

B2B Lead Generation Scraper Service (client: Diaz Solutions): Large scale web data extraction and enrichment platform delivering qualified merchant leads to a paying client. Headless browser automation, reverse engineered hidden APIs, and Python ETL feeding a 100K+ record database across 9 verticals with multi signal qualification scoring, exported as verified lead lists on a recurring delivery cadence.

Greenlightdate (Dating Web App, client, live): Full stack production app (greenlightdate.com) built for a paying client. Engaged to refactor the codebase and execute a live Replit → Vercel+Cloud Run → Cloudflare Pages migration; now 15% co-owner and sole backend engineer: Vite/React + Express + Drizzle/Postgres, Stripe billing, OpenAPI → typed client codegen, CI hardened with CodeQL + OpenSSF Scorecard.

EDUCATION

- High School Diploma: Brentwood High School, Brentwood, NY

CERTIFICATIONS

- CompTIA Security+: ID# COMP001021119756 (renewed 2026, expires Feb 2029)